Jasmeet Singh

Robotics Systems | Simulation | Autonomy Engineer

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Work Experience

- **Eight Vectors** [] — *November 2025 - May 2026*
  Robotics Engineer (Contractor) — Remote
  - Led the development of a pilot-ready **ROS 2**-based Intelligent Driver Assistance System (iDAS) for industrial MHEs (forklifts, reach trucks, order pickers).
  - Led the development of a SLAM-based Real-Time Location Tracking System (RTLS) using 3D LiDAR, stereo cameras, and **RTAB-Map**.
  - Developed a predictive collision monitoring system integrated with **NVIDIA Isaac ROS Nvblox** for dynamic environmental cost representation and obstacle awareness.
  - Built a cloud-enabled **Gazebo** simulation platform for rapid prototyping, testing, and validation of the robotics intelligence stack.
  - Implemented automated **CI/CD** pipelines for integration testing and containerized software deployment using GitHub Actions, headless Gazebo, ROS 2, Docker, and Google Cloud Artifact Registry.
  - Worked on **ROS 1 → ROS 2 migration** for a Zurich-based robotics company (**Anybotics**) as a service provider.

- **Peer Robotics** [] — *October 2023 - July 2025*
  Robotics Engineer — Gurugram, India
  - Contributed significantly to the first successful field deployment of Peer3000, the company's flagship collaborative AMR platform.
  - Designed and implemented **docking, docking safety**, and motion-planning controllers for Peer3000, a steering-based mobile robot for autonomous pallet and trolley transport in industrial settings.
  - Developed a **hybrid drive controller** for Peer3000 capable of runtime switching between **Ackermann steering** and differential-drive behavior using custom forward and inverse kinematics.
  - Built a **Stanley controller**-based navigation module for the Peer3000's **ROS 2 navigation stack**, improving path-tracking accuracy in dynamic environments.
  - Handled **CAD to URDF** conversion workflows for Peer and RM series robots.
  - Created the complete simulation stack in **Gazebo Harmonic**, including custom plugins for pallet lifting, trolley grasping, PLC communication, emergency switches, and charging systems, significantly improving development, testing, and integration workflows across teams.
  - Developed a **camera extrinsic calibration framework** using AprilTags and 2D LiDAR ground truths to enhance sensor alignment on existing AMR platforms.

- **Articulus Surgical** [] — *September 2023 - September 2023*
  Robotics Freelancer — Remote
  - Integrated company's surgical robot and its teleoperating controller with the **microROS** framework.

- **Gazebo, Open Robotics** [] — *June 2023 - August 2023*
  Google Summer of Code (GSoC) — Remote
  - Extended the libsdformat C++ library to **compute mass and moments of inertia** for robot links based on primitive geometries and material densities.
  - Designed a plugin-based architecture enabling **custom mesh-inertia calculators for Gazebo** simulations, improving extensibility and modularity.
  - Implemented a custom mathematical mesh inertia plugin integrated into **gz-sim**, validated through unit and integration tests achieving over 90% coverage.
  - Contributed to upstream open-source **Gazebo** repositories following rigorous **CI/CD** and review workflows.

- **Acceleration Robotics India** [] — *August 2021 - March 2022, November 2022 - July 2023*
  Robotics Intern — Remote

- Developed **camera-sensor plugins** with image-processing pipelines for **Gazebo**, enhancing realism in simulation-based perception tasks.
- Built **terrain** and **weather simulation** modules using real-world elevation and climate data for AGV testing in **Ignition Gazebo**.
- **Migrated** an existing ROS 1 + Classic Gazebo stack to ROS 2 + Ignition Gazebo, implementing SLAM (**slam-toolbox**) and navigation (**Nav2**).
- Created a **VR-based control interface** for real and simulated **UR5** robotic arms using **Oculus Quest**, ROS, and **MoveIt**, enabling immersive teleoperation experiments.

- **e-Yantra Robotics Lab, IIT Bombay [🌐]** — *June 2022 - July 2022*
  Robotics Intern — Mumbai, India
  - Developed a **ROS**-based mobile robot capable of **SLAM** and autonomous navigation for the eYRC challenge.
  - Designed schematics and **PCBs** for the robot's power system using **KiCAD**, including power distribution, battery management, and Li-ion charging circuits.
  - Built an **ESP32**-based control board integrating IMU, ToF sensors, and motor drivers for onboard control and sensing.

Volunteer Experience

- **Co-Founder, A.T.O.M Robotics Lab** — *November 2020 - Present*
  Maharaja Agrasen Institute of Technology — [🌐]
  - Co-founded **A.T.O.M Robotics Lab**, the institute's first student-led robotics research and development community, introducing a culture of open-source collaboration and project-based learning in robotics.
  - Led the development of several open-source and interdisciplinary projects in autonomous systems, embedded electronics, and perception.
  - Designed and fabricated **MR (ModulaR) Robot** — a custom ROS-based Autonomous Mobile Robot platform with modular attachments for diverse applications such as sanitation, logistics, and material handling.
  - Initiated and maintained **ROS-Perception-Pipeline**, an open-source plug-and-play perception framework providing reusable ROS 2 packages with automated CI/CD testing using GitHub Actions.
  - Initiated industry collaborations with robotics companies and local makerspaces.
  - Fostered a self-sustaining ecosystem of project-based learning, innovation, and open-source development that continues to drive the lab even today.

Education

- **Maharaja Agrasen Institute of Technology, GGSIPU** — *August 2019 - July 2023*
  Bachelor of Technology, Electronics and Communications Engineering — New Delhi, India
  - Graduated in *First Division*. Awarded *Outstanding (O)* grade for *Minor* and *Major* projects.
    Overall CGPA: 8.88/10.00
  - Co-founded *A.T.O.M Robotics Lab*, our institute's first student driven robotics initiative focused on advanced robotics research and development.

Skills

- **Robotics:** ROS / ROS 2, Nav2, Motion Planning, Controls, NVIDIA Isaac ROS, AMR Navigation and Docking, System Design and Architecture
- **Simulation:** Gazebo, Isaac Sim, Unity 3D, Physics-based Simulation, Cloud-based simulation, CAD to URDF / USD
- **Infrastructure:** CI/CD, Integration Testing, Docker, GitHub Actions, GCP, Linux
- **Programming:** C++, Python, Shell Scripting, Git, C#, Elixir, Kotlin
- **Electronics / Embedded Systems:** NVIDIA Jetson Platforms, Raspberry Pi, Sensor Integration, Arduino, ESP32, PCB Designing (KiCAD) Soldering and Fabrication
- **Mechanical / Hardware:** CAD (Fusion360, Blender), 3D Printing, Rapid Prototyping

Honors and Awards

- **First Prize, Rally to Tally Hackathon** *Sept 2021*
  Mitsubishi America [🌐]
  - Secured **First Prize** (USD 750) for developing *S.T.E.W.A.R.D*, an autonomous outdoor inventory management robot for counting large steel pipes. Project Video.
  - Implemented real-time pipe detection and counting using circle-detection–based computer vision algorithms.
  - Built a complete simulation environment in ROS and Gazebo for proof-of-concept demonstration.
  - Integrated ROS with InfluxDB and Grafana for telemetry visualization and analytics.
- **Project Showcase, NASA SpaceROS Summer Sprint Challenge** *November 2024*
  Official Gazebo Community Meeting [🌐]
  - Our team ranked among the **Top 10 global submissions** in NASA's SpaceROS Summer Sprint Challenge; project showcased at the official Gazebo community meeting. Project Video.
  - Generated 3D terrain models of Martian and Lunar surfaces from DEMs (HiRISE imager, Mars Reconnaissance Orbiter and LOLA imager, Lunar Reconnaissance Orbiter) for high-fidelity robotic simulations.
  - Migrated and integrated a lens-flare plugin in Gazebo to improve camera-based perception realism.
  - Developed weather and daylight simulation plugins to emulate fog, dust storms, and lighting variations.
- **First Runner Up, Nx EVOS Hackathon** *May 2024*
  Network Optix [🌐]
  - Awarded **First Runner-Up** (USD 3000) for developing *G.A.R.U.D*, a geospatial wildlife reconnaissance dashboard integrating UAV data and computer vision. Project Video.
  - Integrated YOLOv8 with the Network Optix Video Platform for real-time animal detection and monitoring.
- **Finalist, e-Yantra Robotics Competition** *March 2022*
  e-Yantra Robotics Lab, IIT Bombay [🌐]
  - Ranked among the **Top 5 teams out of 300** nationwide; awarded a two-month internship at IIT Bombay.
  - Developed *Functional Weeder*, a fleet of custom 3D-printed pick-and-place line-following robots.
  - Implemented grid-based path planning, mapping, auction-based fleet management, and manipulation algorithms in Elixir.
- **Finalist, e-Yantra Robotics Competition** *March 2021*
  e-Yantra Robotics Lab, IIT Bombay [🌐]
  - Ranked among the **Top 7 teams out of 500** nationwide.
  - Developed the autonomy stack for *Sahayak Bot*, a differential-drive mobile platform with a 6-DoF UR5 manipulator.
  - Implemented navigation, SLAM, perception, and task planning modules using ROS, Gazebo, and MoveIt!.
- **Project Showcase, MagPi Magazine** *Issue 86, October, 2019*
  Official Raspberry Pi Magazine [🌐]
  - Featured in **MagPi Magazine** for the project *Real Harry Potter Wand using Computer Vision*.
  - Developed a custom SVM-based handwritten character recognition model deployed on Raspberry Pi 3.
  - Designed detection and tracking for an interactive wand with retroreflective tip via a night-vision camera in real time.